DIGHUM160 - Critical Digital Humanities
Instructor: Tom van Nuenen
Final Project

Project title: /r/AskHistorians. /r/AskHerstorians? Or is it /r/AskWarHistorians?

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Word limit: Between 2500 and 3000 words (excluding code cells, outputs, and references).

In [40]:
```
# set up your environment

import pandas as pd
import json

import os
import pickle
from IPython.display import Image, display
```

Introduction

The community I'm investigating is the /r/AskHistorians subreddit. This is the description of the community on its sidebar:

"The Portal for Public History. Please read the rules before participating, as we remove all comments which break the rules. Answers must be in-depth and comprehensive, or they will be removed. (AskHistorians)"

My position in observing this community is relevant to this discussion (Weingarten). Though I'm not an active user in the community, I have read and learned about the past through the subreddit. In this light, I feel like an outsider, but not completely disjointed because of my background as a second generation immigrant college aged student with a Western English speaking background.

When I read the subreddit's description I immediately thought the community feels more stern than other subreddits. The focus on more elaborate discussions is highlighted front and center. As a student, I'm familiar with scholarly discourse, but encountering it on social platforms feels out of place.

Even within the rules regarding answers, the higher standard of discussion is mentioned (Rules). I first looked through the top posts of the past month/year/all time to get a sense of what the frequenting members of the community liked to talk about.

But why does our perception of history matter?

I'm personally interest in stories told through music. But I've recently read a book, *The Anatomy of Genres* by John Truby, about the various genres existing today and how mediums like books, television, and film in the modern day establish, remix, and build upon these genres. The genres themselves have also grown out of myths in the traditional meaning. But I think actual myths and Roland Barthes' myths are similar in the sense that they tell stories of what has and what is. Looking at what kind of stories exist and how they are told in /r/AskHistorians is important because it shapes our understanding of people and how people communicate in a community.

In [46]:
```
display(Image(filename='img/M20.2Drakon.jpg', width=400))
```



(Theoi Project)

Truby makes a key point of within the myth genres between a male myth and female myth. While male myths are about dividing and conquering, the female myth focuses more on combining and growing (Truby 181). Truby relates this to the warrior-shame culture seen in the *Illiad* vs the guilt culture within the *Odysssey*. From the ten years of fighting to the ten years of getting back home between the two stories, there exists a transition of Odysseus as a "man who goes from fighter to searcher to lover" (Truby 182).

Reddit is heavily catered towards a young adult white American male community, though participation in subreddits differs (Sattelberg). When I first explored the community's top posts, I noticed the focus on war and how the masculine aspects of war lead to it being a more popular topic of discussion in a subreddit in the top 1% of Reddit communities.

Skimming through the top 50 all time posts, I notice a lot of questions about the USA. This ties in with the mentioned fact that the social media platform is mainly made up of American users. There seems to be a wide variety of ideas expressed, but I notice many questions about war and culture - the details of communities fighting each other and why things that they did or didn't evolve the way they did.

Initially, I didn't know how to look into how the community interprets the world, but I thought about the way questions were posed. For instance, I looked at the top ~100 posts of the past month and noticed how of the Five Ws questions, "why" questions were most popular at around 50. Then "what" and "how" questions at around 30. I wondered why the "why's" were most asked - maybe because it's harder to figure out why historical events and things took place compared to who or what took place.

I noticed the community's rule also mentions the Five Ws when creating an in depth answer; responses are approved "if it will help an uninformed reader understand the 'Who', 'What', 'When', 'Where', and of course 'Why', surrounding the topic" (Rules).

But why might femininity and masculinity matter in this community?

How we speak about the past heavily shapes what was important and what is important today. In this sense, communities surrounding the past are relevant with how we see the world right now. Analyzing the /r/AskHistorians subreddit through the gender binary is limiting in itself. Because gender doesn't just exist in two forms, nuance is lost when viewing through this lens.

Roland Barthes' concept of mythology, where cultural beliefs and values are presented as natural and inevitable, provides insight into how second-generation gender bias operates in historical discussions. Myths are regard as "factual system". This is the "back door of Nature" that the functions of myth access (Barthes 130). The natural relationship is given. Language freezes in the work of myths. A more stable state gives the impression of myths being part of the way things are (Baumann 13).

In /r/AskHistorians, certain mythologies about gender roles and historical importance may be unconsciously reinforced. For instance, the myth of the masculine hero or the association of war with masculinity can be seen as signifiers that perpetuate deeper signified concepts of male power and importance in history.

This second-generation gender bias is the seemingly neutral practices and structures that reflect the values of the dominant group, usually men, who have historically shaped organizations or fields ("Second-Generation Gender Bias").

In this context, the community's focus on topics like war, military tactics, and government could unintentionally perpetuate masculine-centric narratives. Like mentioned, the prevalence of discussions about soldiers and war leaders, predominantly men, may reinforce traditional gender roles. The association of war with masculine traits like strength, execution, and prowess could further secure these biases.

Even in seemingly neutral topics, the framing of historical events and moments often defaults to masculine archetypes. This bias might not stem from overt discrimination but rather from the inherited values and practices of a field historically dominated by men. The high frequency of moderator interventions could also potentially mask or inadvertently reinforce these biases. By examining how these historical topics are prioritized and discussed, I hope to better understand how second-generation gender bias subtly shapes my perception of history and, by extension, the current societal values .

Analysis

This computational analysis mainly focuses on the comments data of the posts in the /r/AskHistorians subreddit.

TF-IDF

I began thinking I can use TF-IDF to identify key terms and phrases that characterize different historical topics or time periods discussed in the subreddit. I thought I could also compare TF-IDF scores between questions and answers to see how language use differs.

In [57]:
```
# Retrieving the Dataset
with open('data/preproc_pp_comments.pickle', 'rb') as f:
    trigrams = pickle.load(f)

df = pd.read_csv('data/preproc_comments.csv')
```

In [58]:
```
# Removes all rows that are '[removed]' or '[deleted]'
df = df.loc[~df['pp_text'].isin(['[removed]', '[deleted]' ]),:]

# Selects only rows that have >3 characters in selftext
df = df.loc[df['pp_text'].str.len() > 3]
```

In [59]:
```
# Set a specific seed
sample_seed = 123

# Sample 3 rows with the specified seed
df_sample = df.sample(n=20000, random_state=sample_seed)
df_sample.shape
```

Out[59]:
```
(20000, 11)
```

I came across many times when my computer couldn't handle the size of the data so I decided to use a smaller sample, focusing on the comments

In [61]:
```
from sklearn.feature_extraction.text import TfidfVectorizer

# Settings that you use for count vectorizer will go here
tfidf_vectorizer = TfidfVectorizer(max_df=0.5,
                                   decode_error='ignore',
                                   stop_words='english',
                                   smooth_idf=True,
                                   use_idf=True,
                                   min_df=.01,
                                   max_features=400
                                  )

# Fit and transform the texts
tfidf = tfidf_vectorizer.fit_transform(df_sample['pp_text'])
tfidf
```

Out[61]:
```
<20000x400 sparse matrix of type '<class 'numpy.float64'>'
        with 222911 stored elements in Compressed Sparse Row format>
```

In [62]:
```
# Place TF-IDF values in a DataFrame
tfidf_df = pd.DataFrame(tfidf.todense(), columns=tfidf_vectorizer.get_feature_names_out().ravel())
```

In [63]:
```
tfidf_df.head()
```

Out[63]:

	000	able	actual	actually	add	agree	allies	allowed	america	american	...	world	writing	written	wr
0	0.0	0.0	0.0	0.0	0.0	0.000000	0.00000	0.000000	0.0	0.000000	...	0.0	0.0	0.0	
1	0.0	0.0	0.0	0.0	0.0	0.000000	0.21161	0.195293	0.0	0.159425	...	0.0	0.0	0.0	
2	0.0	0.0	0.0	0.0	0.0	0.000000	0.00000	0.000000	0.0	0.000000	...	0.0	0.0	0.0	
3	0.0	0.0	0.0	0.0	0.0	0.489613	0.00000	0.000000	0.0	0.000000	...	0.0	0.0	0.0	
4	0.0	0.0	0.0	0.0	0.0	0.000000	0.00000	0.000000	0.0	0.000000	...	0.0	0.0	0.0	

5 rows × 400 columns

In [64]:
```
# Resetting index to get the right comment ID
df_sample = df_sample.reset_index(drop=True)
df_sample
```

Out[64]:

	idint	idstr	created	author	parent	submission	body	score	distingui
0	27503600173	t1_cmuxlpp	1418597851	alfonsoelsabio	t3_2patbf	t3_2patbf	I can't offer a complete answer to your questi...	12.0	N
1	26588031815	t1_c7ptsnb	1357152860	Prufrock451	t1_c7psltr	t3_15tpzr	The Axis position in North Africa was necessar...	12.0	N
2	27418366753	t1_clg6r6p	1413943698	Searocksandtrees	t3_2jxuby	t3_2jxuby	hi! there have been loads of questions about t...	13.0	N
3	27140905933	t1_cguzt1p	1397755790	ASAPBULLWINKLE	t1_cgtv8ga	t3_23637z	I agree with you, I was being a little sarcast...	2.0	N
4	27161242128	t1_ch73ok0	1398972417	Bernardito	t1_ch73h9s	t3_24h173	You're absolutely right! Essentially, life car...	1.0	N
...	...	...	...	...	...	...	...	...	
19995	26411673868	t1_c4sttzg	1338095007	IslandGreetings	t1_c4stlhn	t3_u5co1	Nonsense, Christianity was fairly big deal dur...	2.0	N
19996	26597802760	t1_c7vn7yg	1358009340	lukeweiss	t1_c7vn22h	t3_16fdu8	The criminal negligence is the label I use for...	4.0	N
19997	26727175441	t1_ca0o4k1	1368808656	quince23	t1_ca0mdfc	t3_1ehinj	> Anecdotal I know, but do you think the "lead...	3.0	N
19998	27581010182	t1_co50rnq	1422566664	avenlanzer	t1_co4rc5o	t3_2u1w0r	Learn not to read verses, but context. This ha...	1.0	N
19999	26429008766	t1_c535doe	1340210269	pp86	t1_c530yyt	t3_vbdkb	Well I see the biggest reason for collapse of ...	3.0	N

20000 rows × 11 columns

I reset the index to look at some of the individual comments and what the context of the words were.

In [66]:
```
# Subsetting one DF with the mask of another DF
tfidf_someword_df = df_sample[tfidf_df['kind'] > 0.5]
tfidf_someword_df.head(4)
```

Out[66]:

	idint	idstr	created	author	parent	submission	body	score	distinguish
1183	26662467052	t1_c8y578s	1363652676	eirikraudi	t1_c8y3nlr	t3_1ajb6b	that is true.. also Pawn Stars is scripted... ...	2.0	NaN
2001	27228727807	t1_ciba4u7	1403144237	missginj	t3_27wtn0	t3_27wtn0	Interesting question! Some questions for you, ...	1.0	NaN
2153	26643704134	t1_c8mz1om	1362017181	failuer101	t1_c8mcwci	t3_1997b0	your right, i just kind of seperate the plebs ...	1.0	NaN
2658	27377349037	t1_ckrrlpp	1411606815	CzarKurczewski	t1_ckrehvd	t3_2hbygz	Do you know what kind of costs the English, li...	1.0	NaN

In [67]:
```
#example of one of the comments
print(tfidf_someword_df['body'].iloc[0])
```

```
that is true.. also Pawn Stars is scripted... it's just sad really that his is their best regular show wi
th history of any kind...
```

In this example, I looked through a list of feminine and masculine coded words to see if they existed in my sample (UCLA Equity Office). In comparison to the masculine coded words like fight and battle, I had a hard time finding the feminine coded words in my list for TF-IDF. I feel like this relates to the sheer quantity of masculine related comments over feminine coded words.

However, I found that “kind” exists in my dataset. But when I looked for posts with “kind,” the posts used the word for the purpose of the adverb “kind of.” This relates to the importance of the context of words used within a data set.

I also noticed that feminine coded words like agree, understand, response These words interestingly showed a variety of members in the community commenting on the discussions. They used “agree” for approving other people’s comments. “Understand” was used to confirm the comprehension of comments or ask for further explanation. “Response” led to a noticeable frequency of thanking for the responses the post's questions received. I noticed how it affects my view of the community in seeing the positive reception to the interactions. The engagement amongst the members feels more accepting and heartwarming because of it.

TF-IDF Correlations

It was interesting to see the clear separation of the correlations for "men" and "women". I still have an inkling that this speaks more about the association that has existed with activities between men and women though. This form of distant reading on the community didn't get at my interest for understanding second generation gender bias.

In [72]:
```
corr = tfidf_df.corr()
```

In [73]:
```
#women correalted
corr[['men','women']].sort_values(by='women',ascending=False)[:20]
```

Out[73]:

	men	women
women	0.200804	1.000000
men	1.000000	0.200804
children	0.039516	0.082655
culture	0.012388	0.031786
class	0.007795	0.031578
half	0.006502	0.028981
public	0.016381	0.027054
society	0.014425	0.026570
help	0.006214	0.025680
social	0.008282	0.024571
allowed	0.004333	0.020556
family	0.016639	0.020405
control	0.018151	0.019940
life	0.015371	0.018631
law	0.006999	0.018511
outside	0.011345	0.017211
period	0.011699	0.016965
extremely	0.005839	0.016536
white	0.020479	0.016152
possible	-0.003518	0.015684

In [74]:
```
#men correalted
corr[['men','women']].sort_values(by='men',ascending=False)[:20]
```

Out[74]:

	men	women
men	1.000000	0.200804
women	0.200804	1.000000
army	0.072584	0.001786
soldiers	0.067942	0.006092
battle	0.065060	-0.008700
enemy	0.054405	-0.004416
man	0.054171	0.014917
line	0.049166	0.008653
fighting	0.049051	-0.008523
killed	0.046645	-0.001946
troops	0.044069	-0.006490
armies	0.042813	-0.001795
000	0.039729	-0.003481
children	0.039516	0.082655
war	0.038772	-0.001198
home	0.037674	0.012254
fight	0.035830	-0.007397
particularly	0.032865	0.008320
great	0.030537	-0.002895
number	0.030361	-0.000917

Topic Modeling

Going back to one of my first notices from reading the subreddit, I saw topic modeling as a useful method to see if military related history was actually popular or not. I still wanted to create a new topic model visualization that could run in this notebook though I use a separate topic modeling from before because of hardware limitations.

In [77]:
```
#Loading in the data
df = pd.read_csv('data/preproc_comments.csv')
```

In [78]:
```
# Removes all rows that are '[removed]' or '[deleted]'
df = df.loc[~df['pp_text'].isin(['[removed]', '[deleted]' ]),:]

# Selects only rows that have >3 characters in selftext
df = df.loc[df['pp_text'].str.len() > 3]
```

In [79]:
```
df.shape
```

Out[79]: (49836, 11)

In [80]:
```
from tqdm import tqdm

lemmas_split = [lemma.split() for lemma in tqdm(df['pp_text'])]
```

```
100%|██████████| 49836/49836 [00:00<00:00, 333070.42it/s]
```

In [81]:
```
#Creating a Dictionary from gensim
from gensim import corpora, models, similarities
from gensim.models.coherencemodel import CoherenceModel

# Creates Dictionary
dictionary = corpora.Dictionary(tqdm(lemmas_split))

# filters extremes and assign new ids
dictionary.filter_extremes(no_below=10, no_above=0.4)
dictionary.compactify()

# SAVE DICT
dictionary.save('data/preproc_comments_2.dict')

# Creates Document-Term Matrix of our whole corpus
corpus = [dictionary.doc2bow(text) for text in tqdm(lemmas_split)]
```

```
100%|██████████| 49836/49836 [00:01<00:00, 36445.06it/s]
100%|██████████| 49836/49836 [00:00<00:00, 55485.49it/s]
```

In [82]:
```
from gensim.models.ldamodel import LdaModel

%time
lda_model = LdaModel(corpus=tqdm(corpus),   # stream of document vectors or sparse matrix of shape
            id2word=dictionary,       # mapping from word IDs to words (for determining vocab size)
            num_topics=6,            # amount of topics
            random_state=100,         # seed to generate random state; useful for reproducibility
            passes=2,                 # amount of iterations/epochs
            per_word_topics=False)    # computing most-likely topics for each word
```

```
CPU times: user 1 µs, sys: 1 µs, total: 2 µs
Wall time: 2.86 µs
100%|██████████| 49836/49836 [00:07<00:00, 6356.24it/s]
```

In [83]:
```
%pip install pyLDAvis
```

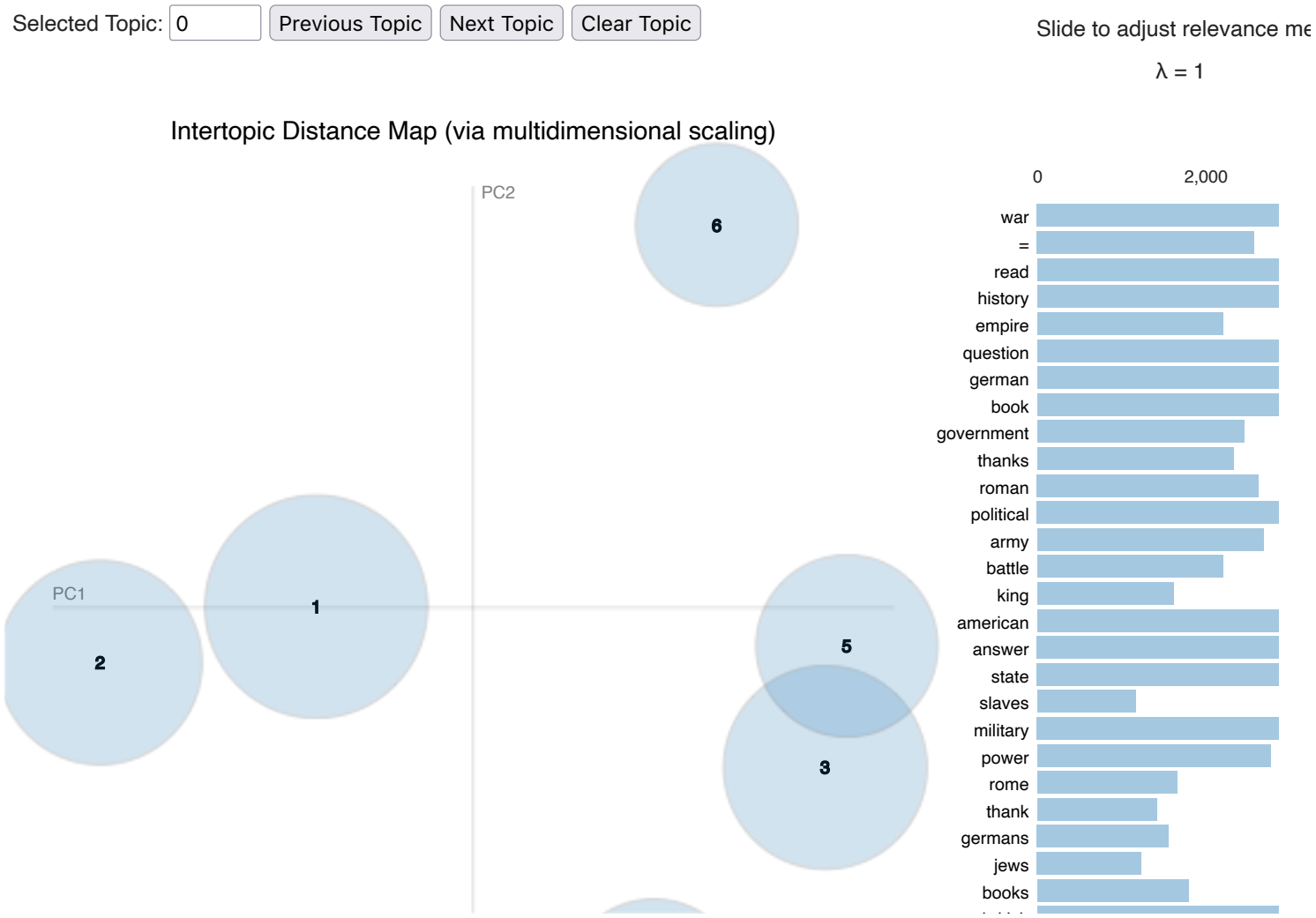
```
Requirement already satisfied: pyLDAvis in /opt/anaconda3/lib/python3.12/site-packages (3.4.1)
Requirement already satisfied: numpy>=1.24.2 in /opt/anaconda3/lib/python3.12/site-packages (from pyLDAvi
s) (1.26.4)
Requirement already satisfied: scipy in /opt/anaconda3/lib/python3.12/site-packages (from pyLDAvis) (1.1
2.0)
Requirement already satisfied: pandas>=2.0.0 in /opt/anaconda3/lib/python3.12/site-packages (from pyLDAvi
s) (2.2.2)
Requirement already satisfied: joblib>=1.2.0 in /opt/anaconda3/lib/python3.12/site-packages (from pyLDAvi
s) (1.4.2)
Requirement already satisfied: jinja2 in /opt/anaconda3/lib/python3.12/site-packages (from pyLDAvis)
(3.1.4)
Requirement already satisfied: numexpr in /opt/anaconda3/lib/python3.12/site-packages (from pyLDAvis)
(2.8.7)
Requirement already satisfied: funcy in /opt/anaconda3/lib/python3.12/site-packages (from pyLDAvis) (2.0)
Requirement already satisfied: scikit-learn>=1.0.0 in /opt/anaconda3/lib/python3.12/site-packages (from p
yLDAvis) (1.4.2)
Requirement already satisfied: gensim in /opt/anaconda3/lib/python3.12/site-packages (from pyLDAvis)
(4.3.2)
Requirement already satisfied: setuptools in /opt/anaconda3/lib/python3.12/site-packages (from pyLDAvis)
(69.5.1)
Requirement already satisfied: python-dateutil>=2.8.2 in /opt/anaconda3/lib/python3.12/site-packages (fro
m pandas>=2.0.0->pyLDAvis) (2.9.0.post0)
Requirement already satisfied: pytz>=2020.1 in /opt/anaconda3/lib/python3.12/site-packages (from pandas>=
2.0.0->pyLDAvis) (2024.1)
Requirement already satisfied: tzdata>=2022.7 in /opt/anaconda3/lib/python3.12/site-packages (from pandas
>=2.0.0->pyLDAvis) (2023.3)
Requirement already satisfied: threadpoolctl>=2.0.0 in /opt/anaconda3/lib/python3.12/site-packages (from
scikit-learn>=1.0.0->pyLDAvis) (2.2.0)
Requirement already satisfied: smart-open>=1.8.1 in /opt/anaconda3/lib/python3.12/site-packages (from gen
sim->pyLDAvis) (5.2.1)
Requirement already satisfied: MarkupSafe>=2.0 in /opt/anaconda3/lib/python3.12/site-packages (from jinja
2->pyLDAvis) (2.1.3)
Requirement already satisfied: six>=1.5 in /opt/anaconda3/lib/python3.12/site-packages (from python-dateu
til>=2.8.2->pandas>=2.0.0->pyLDAvis) (1.16.0)
Note: you may need to restart the kernel to use updated packages.
```

In [84]:
```
import pyLDAvis
import pyLDAvis.gensim_models as gensimvis
pyLDAvis.enable_notebook()

# feed the LDA model into the pyLDAvis instance
lda_viz = gensimvis.prepare(lda_model, corpus, dictionary)
lda_viz
```

```
/opt/anaconda3/lib/python3.12/site-packages/joblib/externals/loky/backend/fork_exec.py:38: DeprecationWar
ning: This process (pid=59472) is multi-threaded, use of fork() may lead to deadlocks in the child.
  pid = os.fork()
/opt/anaconda3/lib/python3.12/site-packages/joblib/externals/loky/backend/fork_exec.py:38: DeprecationWar
ning: This process (pid=59472) is multi-threaded, use of fork() may lead to deadlocks in the child.
  pid = os.fork()
/opt/anaconda3/lib/python3.12/site-packages/joblib/externals/loky/backend/fork_exec.py:38: DeprecationWar
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  pid = os.fork()
/opt/anaconda3/lib/python3.12/site-packages/joblib/externals/loky/backend/fork_exec.py:38: DeprecationWar
ning: This process (pid=59472) is multi-threaded, use of fork() may lead to deadlocks in the child.
  pid = os.fork()
/opt/anaconda3/lib/python3.12/site-packages/joblib/externals/loky/backend/fork_exec.py:38: DeprecationWar
ning: This process (pid=59472) is multi-threaded, use of fork() may lead to deadlocks in the child.
  pid = os.fork()
/opt/anaconda3/lib/python3.12/site-packages/joblib/externals/loky/backend/fork_exec.py:38: DeprecationWar
ning: This process (pid=59472) is multi-threaded, use of fork() may lead to deadlocks in the child.
  pid = os.fork()
/opt/anaconda3/lib/python3.12/site-packages/joblib/externals/loky/backend/fork_exec.py:38: DeprecationWar
ning: This process (pid=59472) is multi-threaded, use of fork() may lead to deadlocks in the child.
  pid = os.fork()
/opt/anaconda3/lib/python3.12/site-packages/joblib/externals/loky/backend/fork_exec.py:38: DeprecationWar
ning: This process (pid=59472) is multi-threaded, use of fork() may lead to deadlocks in the child.
  pid = os.fork()
```

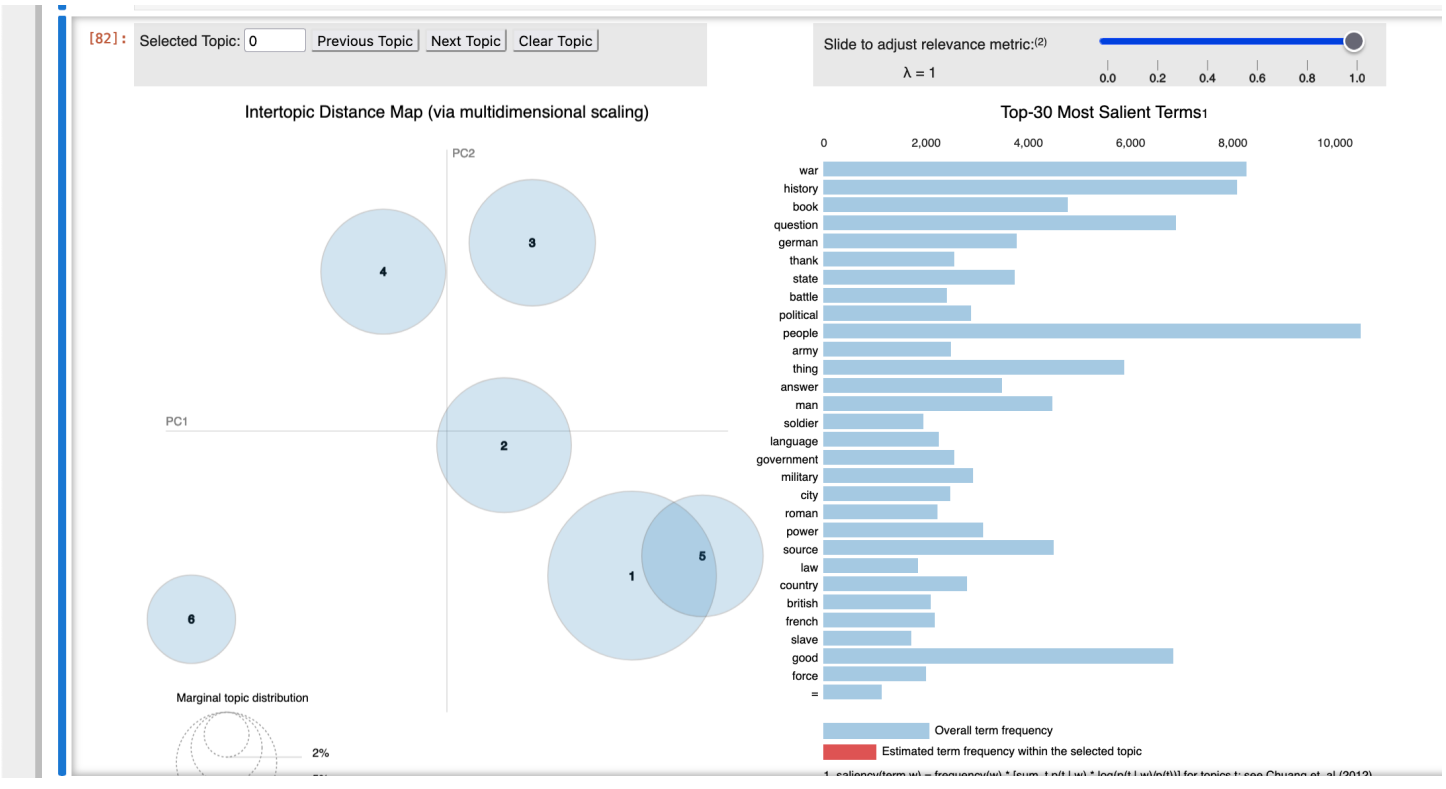
Out[84]:



In this example of topic modeling on the data, war is prevalent in topics 3, 4, and 5. Topic 3 seems modern with the terms soviet and allies. Topic 4 seems a bit older with the mentions of american, revolution, and civil_war. And topic 5 looks to be even older with the frequency of terms like king, empire, and roman.

In [86]:

```
display(Image(filename='img/t30_terms_6topic.png', width=800))
```



This is my initial experience with topic modeling. I've used topic modeling to create a broad 6 topics of the community:

Religion, Culture

Meta

Military Tactic, Technology

Military Operation, Strategy

Time, Period

Government, Law

These were labeled by reading through the top 10-20 comments under each topic. Maybe because I was looking for the masculine feminine relationship, I saw the two military topics to be strongly about military and its different levels of war: strategic, operational, tactical (Harvey). Unlike the current example, my first topic separation only had two distinct military related categories. This could indicate a even greater prevalence of war time related conversations throughout the subreddit.

The biggest concern is the majority of the comments are under Meta. My guess is that the majority of the comments in this topic are moderator comments explaining why the comment it is replying to was deleted.

In [88]:
```
#loading the topic labeled data
pos_lemmas_topics_df = pd.read_csv('data/preproc_comments_pos_lemmas_topics.csv')
```

In [89]:
```
def get_percent_distribution(df, column_name):
  """
  Calculates the percentage distribution of rows with unique values in a column.
  """

  # Get the counts of unique values
  value_counts = df[column_name].value_counts()

  # Calculate the total number of rows
  total_rows = len(df)

  # Calculate the percentage distribution
  percent_distribution = (value_counts / total_rows) * 100

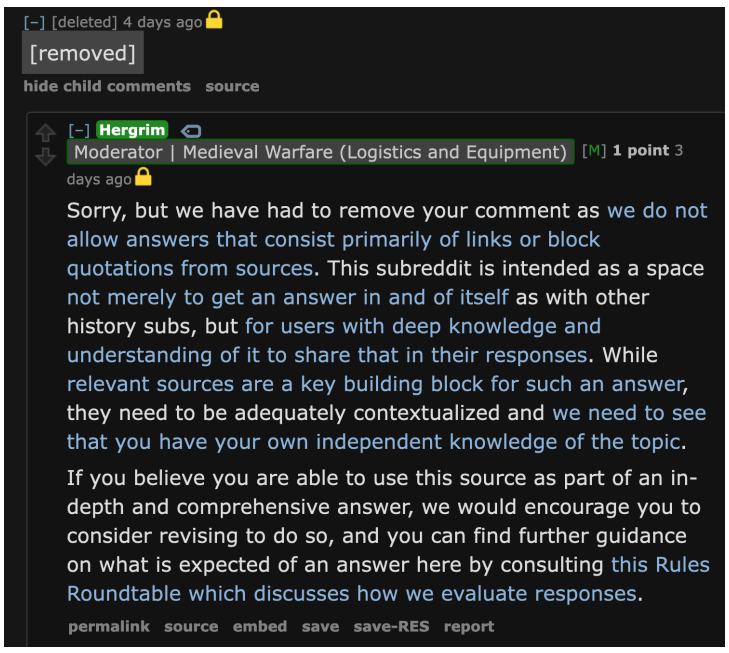
  return percent_distribution
percent_distribution = get_percent_distribution(pos_lemmas_topics_df, 'dom_topic')
print(percent_distribution)
```

```
dom_topic
Meta                           58.869091
Time, Period                   11.951200
Military Operation, Strategy    9.276427
Government, Law                 8.435669
Religion, Culture               7.863793
Military Tactic, Technology     3.274741
Name: count, dtype: float64
```

Though there are many individual comments in the Meta topic, the comments are frequently moderator level responses to deleted comments. Seeing rows of [removed] in many posts while browsing the subreddit led to this perspective.

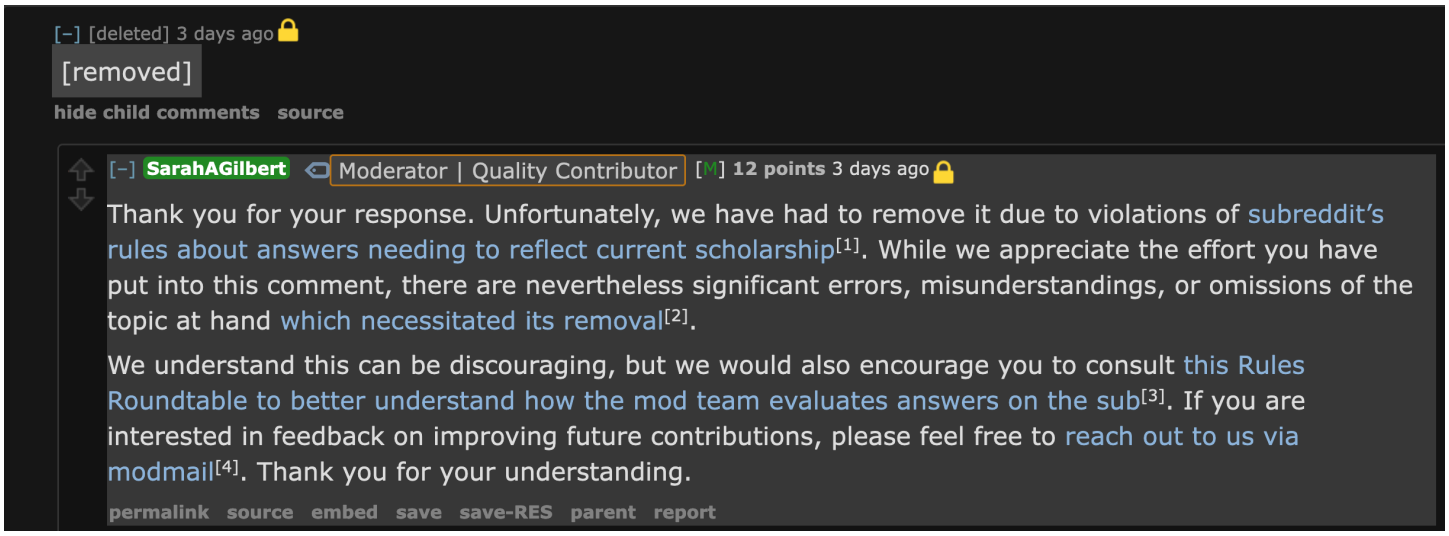
These are some examples of deleted comments with moderator replies that outline the reasons why the comment was removed.

In [92]:
```
display(Image(filename='img/Image of Discussion on Communism and Egalitarianism.png', width=400))
```



(Image of Discussion on Communism and Egalitarianism)

In [94]:
```
display(Image(filename='img/Image of Discussion on Woodrow Wilson.png', width=800))
```



(Image of Discussion on Woodrow Wilson)

Through topic modeling I saw that a traditionally masculine topic was actually as dominant as I thought. But I still didn't feel like I was getting at the root of my research of foundational gender bias that existed in how the community spoke about these topics of history. Through the topic labeling, I decided to analyze a close reading of some of the comments under the two military labeled topics.

Sample Comment Close Reading

```
In [98]: display(Image(filename='img/strange_rule_Military Operation_Strategy.png', width=900))
```

```
[371]: filtered_df['body'][17856]
```

```
[371]: 'The short answer is yes, they had strange and bizarre rules that seem utterly alien to anyone not expected to operate in a combat zone, ju
st as contemporary militaries have strange and bizarre rules, which again seem insane when viewed by civilians.\n\nWith a bit more investig
ation, you\'ll find that while these rules appear to be strange and bizarre, the vast majority of them have some purpose, albeit the purpos
e is often misunderstood or not understood, period.\n\nTo paraphrase my own Drill Instructor, everything that the military has a recruit do
has some purpose behind it.  Just because the purpose is not immediately understood does not mean that it is useless.  To be fair, however,
this *is* sometimes the case.\n\nI\'ll finish with two examples, one that is strange and bizarre, but ultimately has a purpose.  The second
is strange and bizarre, and ultimately served no purpose.\n\n-At Parris Island, and I\'ll assume San Diego, Marine recruits are issued a ca
nteen, and instructed to hold the canteen at all times in their left hand, with the arm on the outside of the canteen, with the arm extende
d straight along the side of one\'s torso.  This appears to be both strange and bizarre, until one is issued a rifle.  Suddenly it becomes
clear that the days you\'ve been holding your canteen in an awkward manner has magically increased your muscle strength for when you are is
sued a rifle, and instructed to hold it in the exact same manner as that which you held your canteen.\n\n-Example number two illustrates wh
at I believe you are implying with your question, that sometimes this devotion to strange and bizarre training methods results in less than
optimal results.  During World War II, the British Army, for obvious reasons, was looking for ways to streamline all of their methods of tr
aining.  They submitted video footage of an artillery crew in action to an efficiency expert.  After several days of reviewing the footage,
the expert came back with a question.  He noticed that at a point right before the cannons were fired, one of the artillery crewmen would s
tand back, extend both his arms downward in awkward manner, and hold them there until the gun had been fired.  The Army officers were flumm
oxed by this admittedly strange and bizarre step, that was indeed present in all the training manuals.  It took several days, however event
ually a retired Sergeant Major who bordered on the ancient was found who was able to explain the step.  After viewing the footage, he turne
d to both the officers and efficiency expert, and explained that the man was doing that movement in order to "hold the horses and keep them
from running off."  The movement was immediately removed from future editions of the manual.\n\n*EDIT:  On a final point, I would say that
if you find yourself chafing under all these "strange and bizarre" rules, you are not alone, and should not feel that there is something wr
ong with you.  You are simply retaining some remnants of a civilian outlook.  These will either slowly erode away, or they will not, in whi
ch case I doubt you are much of a candidate for reenlistment.  I\'ll leave you with Rudyard Kipling\'s [The \'eathen](http://www.daypoems.n
et/poems/1870.html), which might help you deal with your transformation.  Goodness knows it helped me deal with it.* \n\n'
```

(Example 1)

This comment focuses on “strange”, “bizarre”, and “alien” practices that happen in the military. The weird training methods are ultimately to build strength and resilience. These qualities traditionally seen as masculine are shaped as the main qualities of a soldier; ones that a civilian would have to adapt to. Though the commenter adds that there isn’t “something wrong with you” if you do feel annoyed by the weird rules, they describe the overcoming of masculine traits as “erod[ing] away” the “civilian outlook.” The language underlines a foreign outlook on masculinity where non soldiers need to suppress their non masculine traits to take part in the military. I couldn’t find the comment’s thread because the account was suspended, but I think the comment itself is an example of masculine normativity in conversations about topics covering war and military.

In [101…
```
display(Image(filename='img/sniper_comment_Military_tactics_Technology.png', width=900))
```

[466]:
```
sorted_df['body'][45120]
```

[466]:
```
'To piggy back on /u/mosin91, the idea of a sniper comes from the use of light infantry as a result of the Frederick the Great. While I can
not speak much about Frederick\'s light infantry, I can explain the purpose.\n\nDuring the Seven Years War, the line regiment reigned supre
me. Men were packed together in lines to ensure that volley fire could cause damage to the enemy infantry due to the inaccuracy of the musk
et. In order to combat this, light infantry would march a little bit ahead of the regular line infantry and fight in open order where men w
ere at least an arms width apart from each other. Here they could hide behind whatever they found as well as crouch to protect themselves w
hile they fire and reload.\n\nThe purpose of light infantry was to use a handful of skilled sharpshooters to shoot at the enemy line infant
ry that was clumped together. While they don\'t have the raw shock power of the line, they would be able to have more of an effect since th
ey would have time to reload and aim at the enemy, most likely hitting them due to the time they take to reload and aim.\n\nThe French saw
this and eagerly took it up in theory in the interwar period of the American Revolution and the French Revolution. It comes to a head when
the French would use skirmishers in a "cloud" to go forward of the relatively untrained recruits in order to poke holes in the enemies line
. However, from reading that I have done, it hasn\'t been proven nor disprove that it happened. Most likely it would have been done at the
lower level of command (most likely a regimental commander or a battalion commander) where soldiers would be thrown forward to prep the ene
my for a column attack.\n\nFrom this, Napoleon would refine the French army to include a specific voltiguer regiment in every battalion of
infantry (be it light or line) of the French army, so that the French had a 1:5 ratio of skirmishers at full strength.'
```

(Example 2)

In this example I found under the Military Tactics and Technology topic, the original post about snipers originating from the use of light infantry as a result of Frederick the Great. The comment discusses the organization of military units like regiments and battalions highlighting masculine order and command. It also displays the importance of progress through the evolution of line infantry to light infantry and skirmishers: The “French would use skirmishers in a ‘cloud’ to go forward of the relatively untrained recruits in order to poke holes in the enemies line.” The focus of masculine improvement for effective fighting is expressed alongside masculine aggression in the ability to “poke holes.”

Word Embeddings

Word embeddings were used to further see the relationship with the traditional masculine and feminine relationship from a wider scale than the example comments. Here I also loaded my previous model from my initial testing of the tool.

In [106…
```
# Package imports
import os
import pandas as pd
import numpy as np

import pickle
from gensim.models import Word2Vec
import multiprocessing

import matplotlib.pyplot as plt
from sklearn.manifold import TSNE

import bokeh
from bokeh.plotting import figure, show, output_notebook
from bokeh.models import HoverTool, ColumnDataSource, LabelSet

output_notebook()
bokeh.io.output_notebook()
```

Loading BokehJS ...

Loading BokehJS ...

In [107…
```
#just to make sure, topic labeled dataframe is reloaded
df = pd.read_csv('data/preproc_comments_pos_lemmas_topics.csv')
```

In [108…
```
df.shape
```

Out[108… (49836, 14)

In [109…
```
#loading model
tsne_path = 'data/tsne_model'
tsne_df_path = 'data/tsne_df.pkl'

with open(tsne_path, 'rb') as f:
    X_tsne = pickle.load(f)

tsne_df = pd.read_pickle(tsne_df_path)
```

Language Biases

In [111…
```
# Import function to calculate biased words
```

```
from utils import calculate_biased_words
```

```
[nltk_data] Downloading package averaged_perceptron_tagger to
[nltk_data]     /Users/a--/nltk_data...
[nltk_data]   Package averaged_perceptron_tagger is already up-to-
[nltk_data]       date!
[nltk_data] Downloading package vader_lexicon to
[nltk_data]     /Users/a--/nltk_data...
[nltk_data]   Package vader_lexicon is already up-to-date!
```

In [112…
```
#target1 = ['general', 'soldier', 'weapon', 'attack', 'victory', 'battle', 'command', 'conquer', 'armor',
#target2 = ['nurse', 'civilian', 'refugee', 'aid', 'negotiate', 'survive', 'rebuild', 'home_front', 'fami

#target1 = ['allies' , 'victory' , 'command' , 'necessary' , 'achieve' , 'manpower' , 'tactical' , 'goal
#target2 = ['sister' , 'female' , 'woman' , 'girl' , 'daughter' , 'she' , 'hers' , 'her']

#target1 = ['strength', 'power', 'aggression', 'strategy', 'leadership', 'conquest', 'heroism', 'courage
#target2 = ['nurture', 'empathy', 'cooperation', 'peace', 'support', 'healing', 'emotion', 'compassion',

#target1 = ['brother', 'male', 'man', 'boy', 'son', 'he', 'his', 'him', 'father', 'king']
#target2 = ['sister', 'female', 'woman', 'girl', 'daughter', 'she', 'hers', 'her', 'wives', 'queen']

target1 = ['analyze', 'conquest', 'dominion', 'challenge', 'categorize', 'transformation into legend', '

target2 = ['birth', 'growth', 'maturity', 'decline', 'death', 'resurrection as a god', 'rebirth', 'repet
```

Here is the target sets I have in terms of Truby's female and male myths. I created target1 based on the "female cyclical beats work through the steps of the growth cycle" as well as words combine and grow (Truby 180). The other commented sets were used before to look at the distinction of some masculine / feminine tropes.

In [114…
```
model = Word2Vec.load('data/embeddings.emb')
```

In [115…
```
[b1, b2] = calculate_biased_words(model, target1, target2, 4)
```

In [116…
```
print('Biased words towards target set 1')
print([word for word in b1.keys()])
```

```
Biased words towards target set 1
['invasion', 'attack', 'spain', 'territory', 'independent', 'conquered', 'interests', 'invaded', 'defeate
d', 'attacked', 'ottomans', 'territories', 'join', 'alliance', 'hannibal', 'unified', 'invade', 'neutral'
, 'sovereignty', 'intervention', 'decide', 'invading', 'secure', 'defending', 'hoped', 'franco', 'confede
ration', 'anglo']
```

In [117…
```
print('Biased words towards target set 2')
print([word for word in b2.keys()] )
```

```
Biased words towards target set 2
['low', 'rate', 'increase', 'increased', 'price', 'increasing', 'alcohol', 'beer', 'bread', 'levels', 'he
alth', 'meat', 'rates', 'rising', 'drinking', 'prices', 'percent', 'wages', 'consumption', 'fat', 'milk',
'shoes', 'corn', 'crop', 'wheat', 'explosion', 'consumed', 'baby', 'cent', 'steady', 'infection']
```

In [118…
```
import matplotlib.pyplot as plt
import numpy as np

from sklearn.manifold import TSNE
%matplotlib inline
```

```
/opt/anaconda3/lib/python3.12/site-packages/IPython/core/pylabtools.py:77: DeprecationWarning: backends i
s deprecated since IPython 8.24, backends are managed in matplotlib and can be externally registered.
  warnings.warn(
/opt/anaconda3/lib/python3.12/site-packages/IPython/core/pylabtools.py:77: DeprecationWarning: backend2gu
i is deprecated since IPython 8.24, backends are managed in matplotlib and can be externally registered.
  warnings.warn(
```

In [119…
```
with open(tsne_path, 'rb') as f:
    X_tsne = pickle.load(f)

tsne_df = pd.read_pickle(tsne_df_path)
```

In [120…
```
# Convert biased term keys to arrays
target1_idx = np.array([model.wv.key_to_index[key] for key in b1.keys()])
target2_idx = np.array([model.wv.key_to_index[key] for key in b2.keys()])
```

In [121…
```
# Find t-sne values for the biased sets
X_target1 = X_tsne[target1_idx]
X_target2 = X_tsne[target2_idx]
```

In [122…
```
from bokeh.io import show, output_notebook, output_file
from bokeh.plotting import figure
```

```
from bokeh.models import ColumnDataSource, LabelSet

# Set up the Bokeh plot
output_notebook()

p = figure()

# Create ColumnDataSource for X_target1 (blue)
source1 = ColumnDataSource(data=dict(x=X_target1[:, 0], y=X_target1[:, 1], label=[model.wv.index_to_key[

# Create ColumnDataSource for X_target2 (red)
source2 = ColumnDataSource(data=dict(x=X_target2[:, 0], y=X_target2[:, 1], label=[model.wv.index_to_key[

# Add scatter plot for X_target1 (blue)
p.scatter(x='x', y='y', color='blue', size=8, source=source1)

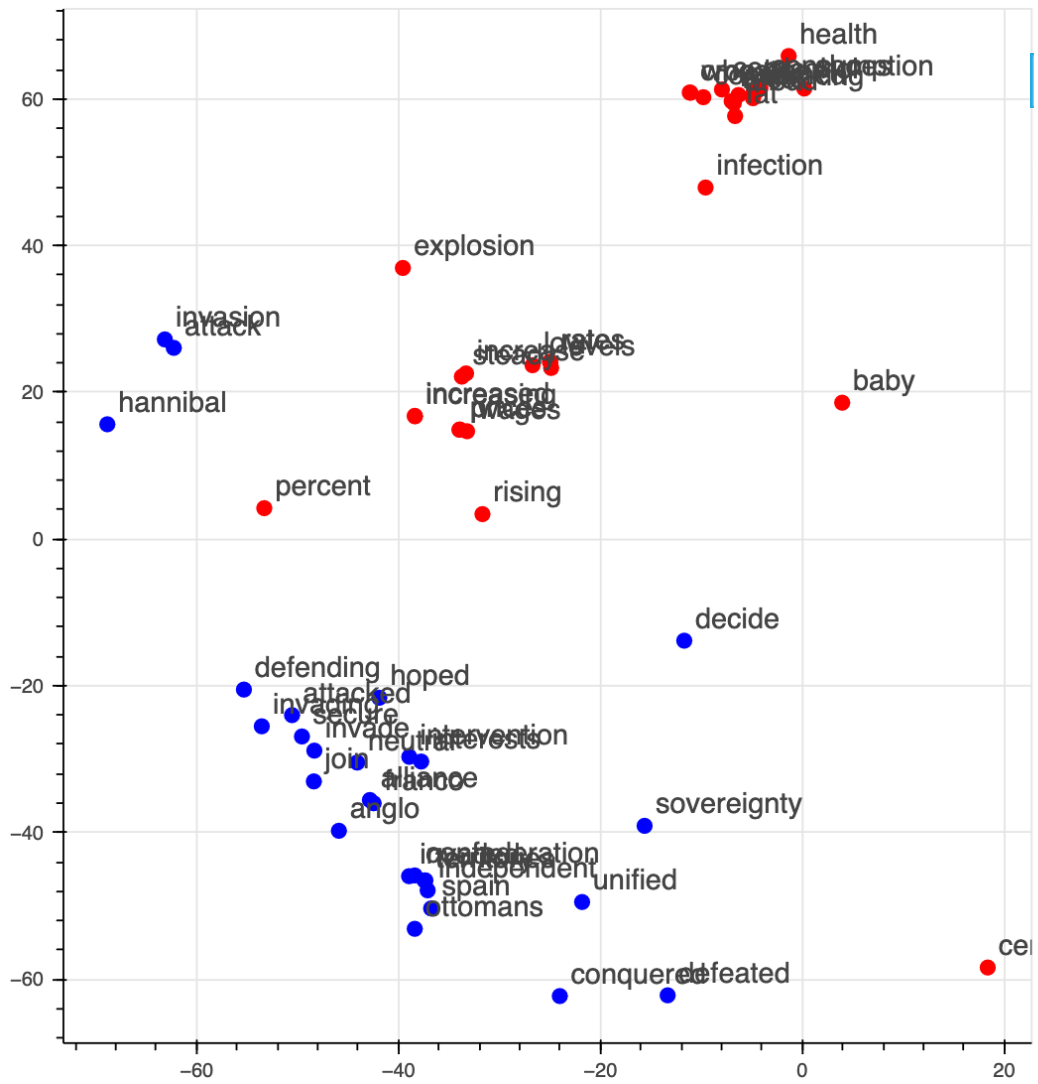
# Add scatter plot for X_target2 (red)
p.scatter(x='x', y='y', color='red', size=8, source=source2)

# Add labels for X_target1
labels1 = LabelSet(x='x', y='y', text='label', x_offset=6, y_offset=3, source=source1)
p.add_layout(labels1)

# Add labels for X_target2
labels2 = LabelSet(x='x', y='y', text='label', x_offset=6, y_offset=3, source=source2)
p.add_layout(labels2)

# Show the plot
show(p)
```

BokehJS 3.4.1 successfully loaded.



In this plot of the Truby target words, the growth cycle words had biased words surrounding health, wealth, and food. Whereas unsurprisingly, the masculine myth set had assertive words around war. I instantly eyed "hannibal" in the upper left hand near "invasion" and "attack." Previously with the other target sets I recall seeing Napoleon aswell. Seeing famous military men led me to think the relationship with singular historic individuals that sum up the masculine traits and qualities is a possiblity in the community.

In my previous explorations on the target sets I found similar observations with the the present distinction between the two groups. Because military history is traditionally based around men, the results of the data show a stronger connection of men to aspects of war compared to women. In the gender binary target set using t-SNE, I noticed women had groups of

self-care, sexuality, and family. On the other hand, men had groups of strategy, campaign, and plans.

I also tried to make my own target set of terms that are traditionally masculine and feminine (like strength vs nurture, aggression vs cooperation, courage vs compassion). The biases result showed a military vs food/economic separation. The word embedding can't easily show masculine and feminine use of language within the topics that are discussed, but it does reinforce the masculine relationship to war that is traditionally seen.

Conclusions

I frequently ran out of ram while working on this final project to bring together the various tools to analyze my data. It is important to acknowledge the limitation of using smaller datasets and looking for relationships that I assume are true and follow the status quo. Recognizing that the data is an old snapshot of the community and what I see is not what exists today is significant.

In my search for how second generation gender bias exists in the /r/AskHistorians community, I found a few instances in the close reading analysis, but the distant reading proved to be more complicated than I believed. But the computational methods still showcase a bias in what the community is generally interested in. In this way, the stories that are told in the subreddit can be seen as masculine leaning in content than form.

The distant reading analysis of the gender binary doesn't easily showcase an underlying masculine/feminine leaning lens of the conversations at hand. The close reading analysis is always a handful of drops in the sea of data that aren't a clear view of what's happening at scale. Thus the analysis is always opaque.

The military emphasis in the community could carry the myth of societal advancement being possible through conflict instead of cooperation. The comment about military parties that focus on overcoming and defeating the enemy reflects the competition and domination.

The military discussions and topics also mainly focus on Western military history. This could reinforce a Eurocentric or Western-centric understanding of global history.

Military technology discussions could also promote technological determinism. It encourages the idea that society is where it is now because of past technological breakthroughs and more specifically warfare technology development.

One area of further research into the data could cover the historians who regularly answer questions and posts. For instance, does the demographic of historians who are specialized in certain topics reflect the research interest of historians today? The community specifies that "the moderator team is looking for responses which are in line with the existing Historiography, on the topic" (Rules). On a larger scale, how has historiography been skewed towards masculine-coded topics?

As I was closing in on this final project I reflected on how my approach towards the data of /r/AskHistorians reflected Truby's "male myth" to divide and conquer the problem / conflict (data / information). I hope that in learning more about how to look conversations in history we can combine and grow.

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Word Count

In [134…

```
# This code cell will give you a word count, in case you need it.
# Note that this opens the file you are currently in, so make sure to save it first to get an accurate w

with open('DIGHUM160_essay_Suh.ipynb',encoding='utf-8') as json_file:
    data = json.load(json_file)

wordCount = 0
for each in data['cells']:
    cellType = each['cell_type']
    if cellType == "markdown":
        content = each['source']
        for line in content:
            # you might want to filter for more markdown keywords here
            temp = [word for word in line.split() if "#" not in word]
            wordCount = wordCount + len(temp)

print(wordCount)
```

```
3068
```

In [ ]: